TDRA Regulations on Unmanned Aircraft Radio Systems (v2.0, 28 Dec 2022)

Telecommunications and Digital Government Regulatory Authority (TDRA)

Regulatory Domain: Telecommunications & C2 Spectrum Regulations

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Article (3.5) - Data Security and Privacy in UAS Wireless Systems

All wireless telemetry, control links, and sensor payloads operated by unmanned aircraft must adhere to UAE federal data protection legislation. C2 link transmissions must implement cryptographic authentication to prevent unauthorized interception or electronic spoofing.

Article (4.1) - Spectrum Allocation for Command & Control (C2)

Unmanned aircraft systems shall operate Command and Control (C2) and payload communications strictly within TDRA-designated spectrum bands: the 5030-5091 MHz aeronautical safety band or authorized commercial 5G NR channels.

Article (4.2) - Mobile Cellular Network Quality of Service (QoS) Thresholds

UAS operating over public cellular infrastructure (4G/5G) must maintain Round-Trip Time (RTT) latency under 100 milliseconds and packet loss below 1.0%. Latency exceeding 250 milliseconds or packet loss above 3.0% constitutes a critical C2 degradation requiring immediate failsafe altitude adjustment or cell handover.